

SCRIBE: Learning Reusable Skill Rubrics for Process Reward Models

Yuxuan Jiang Francis Ferraro
University of Maryland, Baltimore County



[Project & Slides](#)

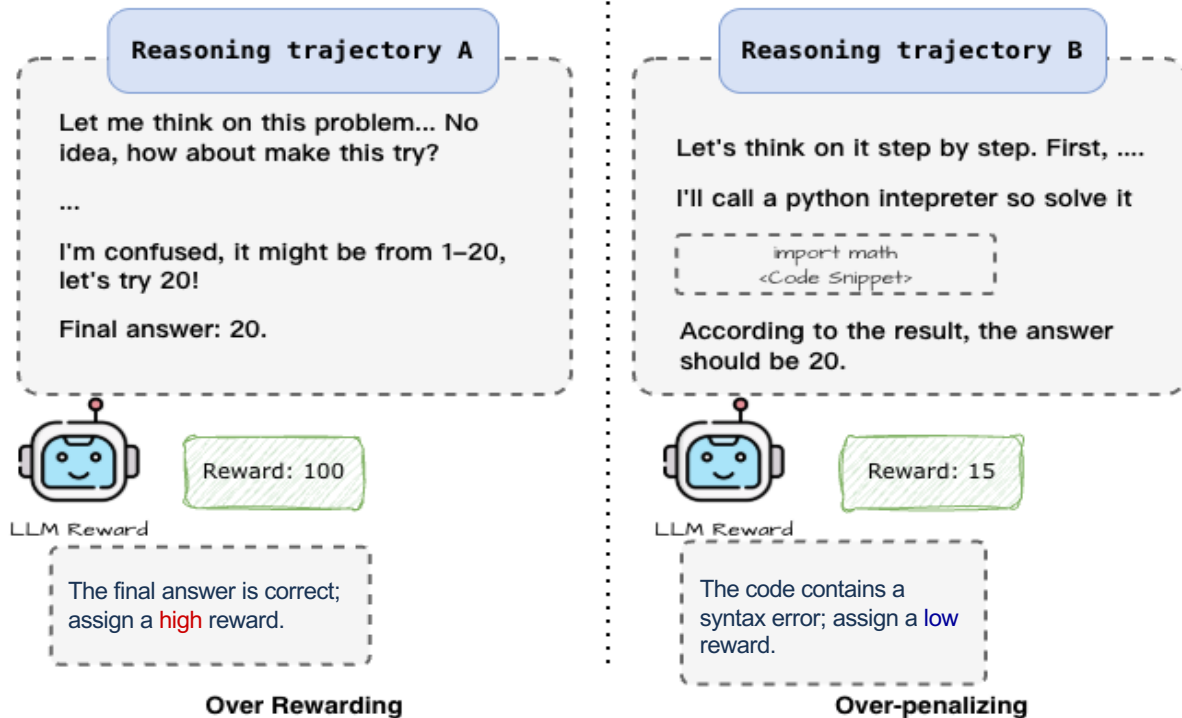


yuxuanj1@umbc.edu

Supported
by:



Unreliable LLM-as-a-Judge Reward



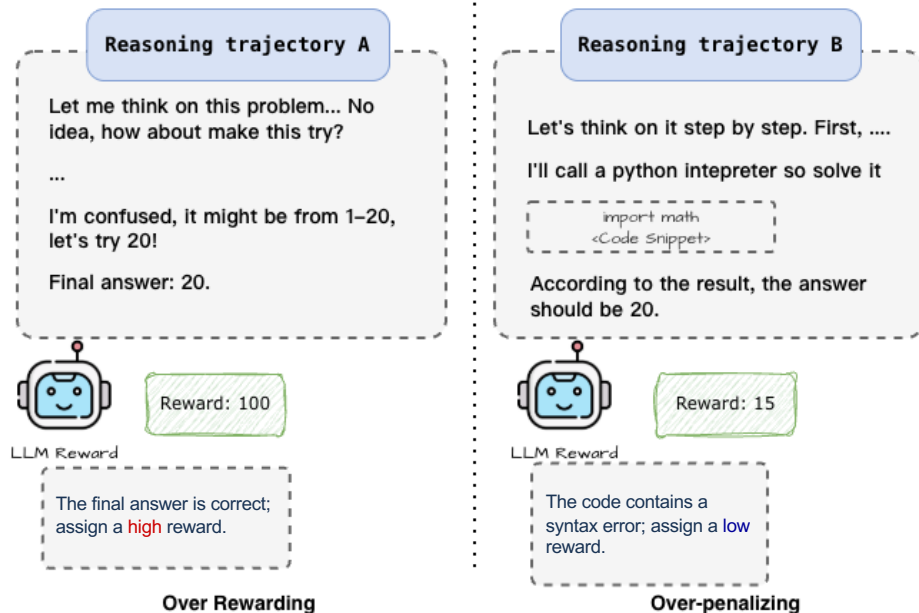
Over-rewarding:
Correct Answer but Flawed Logic

Over-penalizing:
Sound Strategy + Minor Syntax Error

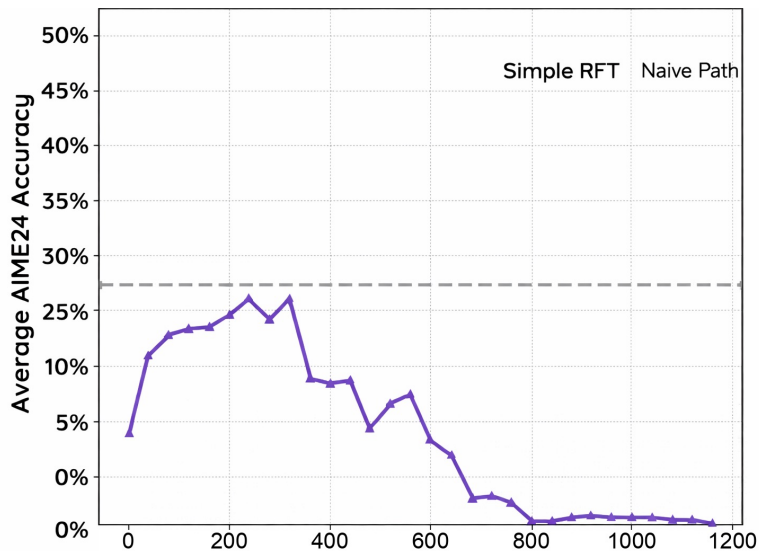
Motivation: Unreliable Reward Leads to Unreliable RL Training

Over-rewarding: Correct Answer + Flawed Logic

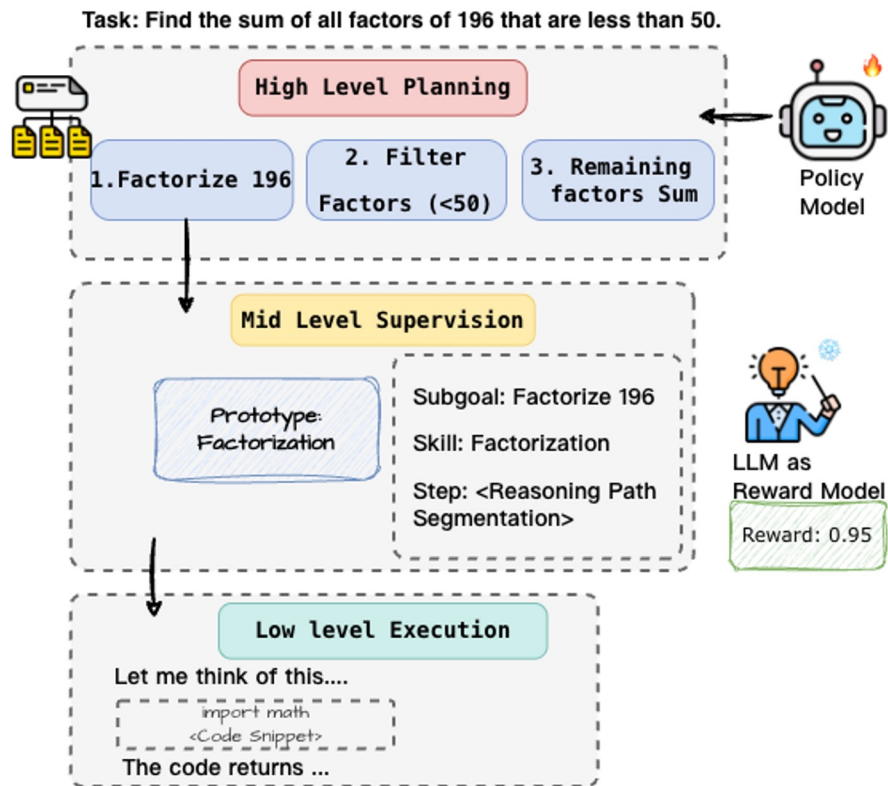
Over-penalizing: Sound Strategy + Minor Syntax Error



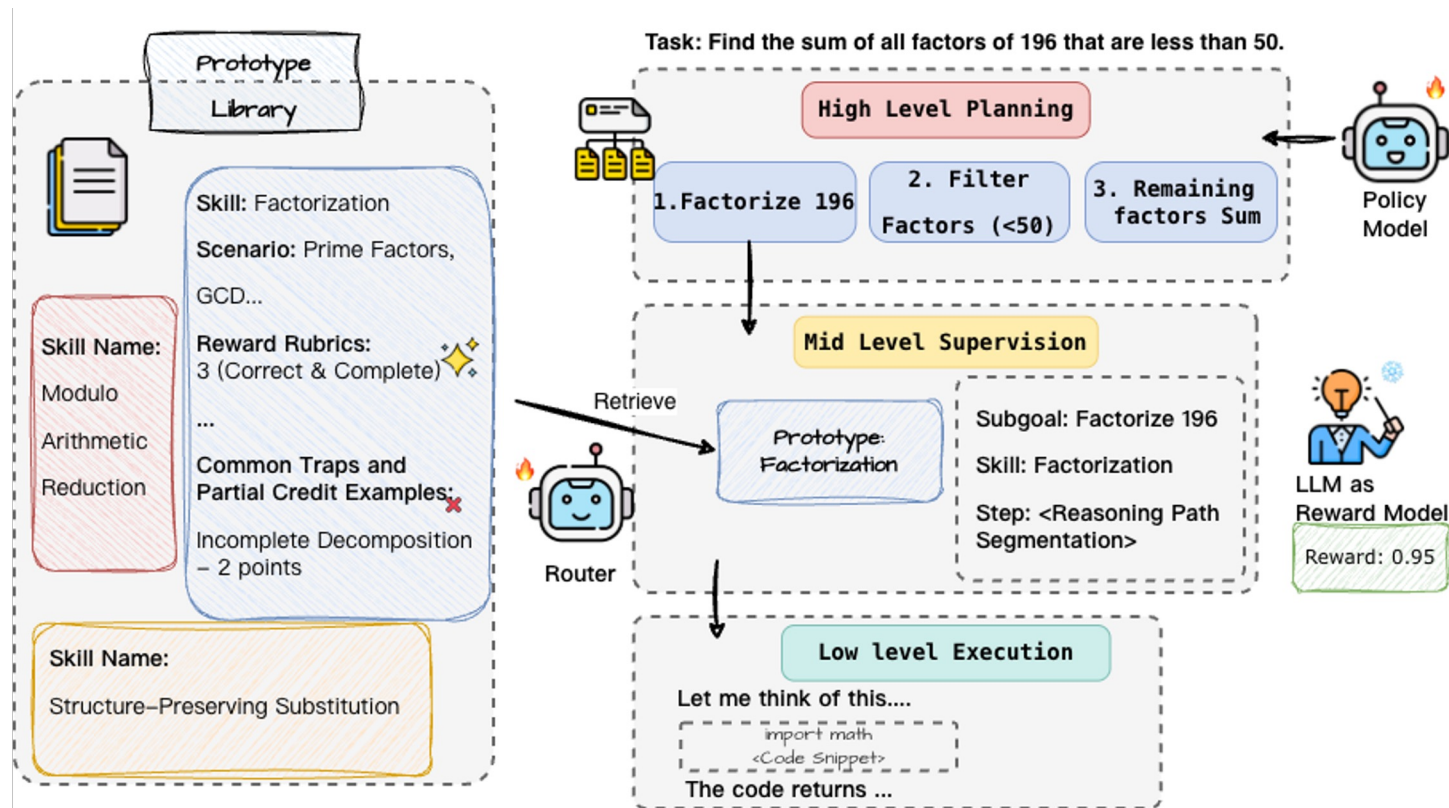
Training Crash!



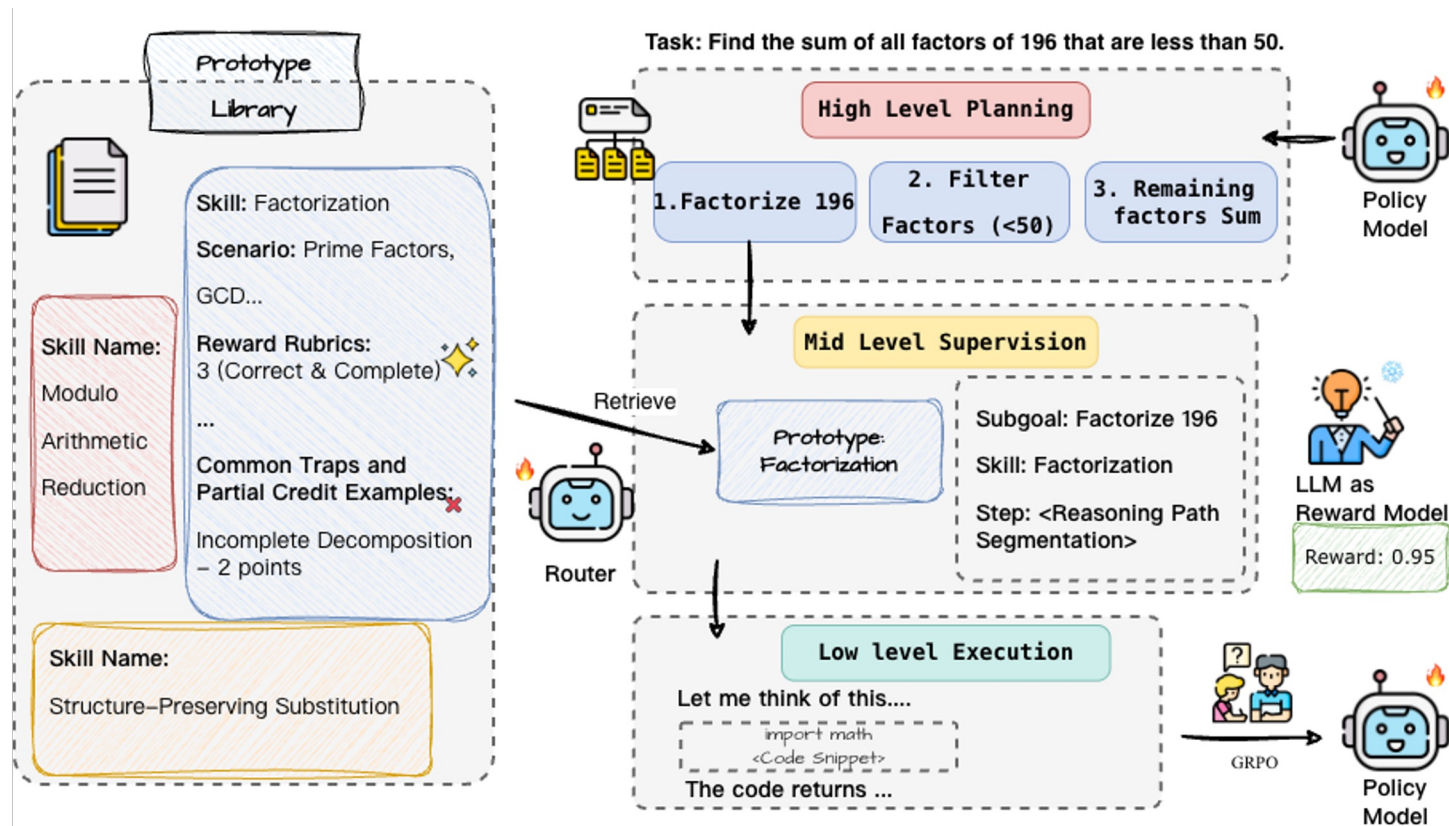
SCRIBE: Decomposing a Goal into Executable Units



Skill Prototype: A Reliable Rewarding Rubric



Reliable Training



A Skill Prototype Becomes a Reusable Reward Standard

SKILL PROTOTYPE · MATHEMATICAL REASONING

Bound-Based Conclusion and Synthesis

SCOPE & APPLICABLE SCENARIO

Bounds, extrema, tightness, case coverage, and domain constraints. Conclude from prior constraints; justify attainability or feasibility.

CANONICAL REASONING PATTERN

1. Identify the relevant intermediate results.
2. Distinguish implication from a tightness claim.
3. Prove attainability when claiming an extremum.
4. Check domains and cover all cases.
5. State the final conclusion.

JUDGING RUBRIC · 0–3

3 Correct & complete: rigorous and feasible.

2 Minor flaw: sound strategy, small slips.

1 Major logical gap: unsupported conclusion.

0 Wrong skill: incompatible or premature.

COMMON TRAPS

Boundary leak → 2 Unproved tightness → 1

TAKEAWAY

Structured

A prototype specifies the skill, expected reasoning pattern, and concrete failure modes.

Reusable

The same rubric evaluates comparable subgoals across different problems.

Actionable

Reward reflects whether an intermediate behavior is complete—not merely whether it resembles a reference.

Reusable criteria reduce reward drift across instances.

Reference Quality Matters for Instance-Level Rubrics

RaR builds a rubric from each problem’s reference answer.
SCRIBE reuses skill rubrics without requiring golden reference answers.

Method	Reference source	MATH500	AIME25	BFCL
RaR	GPT-5.1 labeled	90.7	43.3	46.2
SCRIBE	Not required			

Reference Quality Matters for Instance-Level Rubrics

RaR builds a rubric from each problem’s reference answer.
SCRIBE reuses skill rubrics without requiring golden reference answers.

Method	Reference source	MATH500	AIME25	BFCL
RaR	GPT-5.1 labeled	90.7	43.3	46.2
SCRIBE	Not required	95.8	63.3	51.3

Same Trajectory: Is the Reward Consistent?

Give the judge the same complete trajectory and repeat reward scoring 10 times.



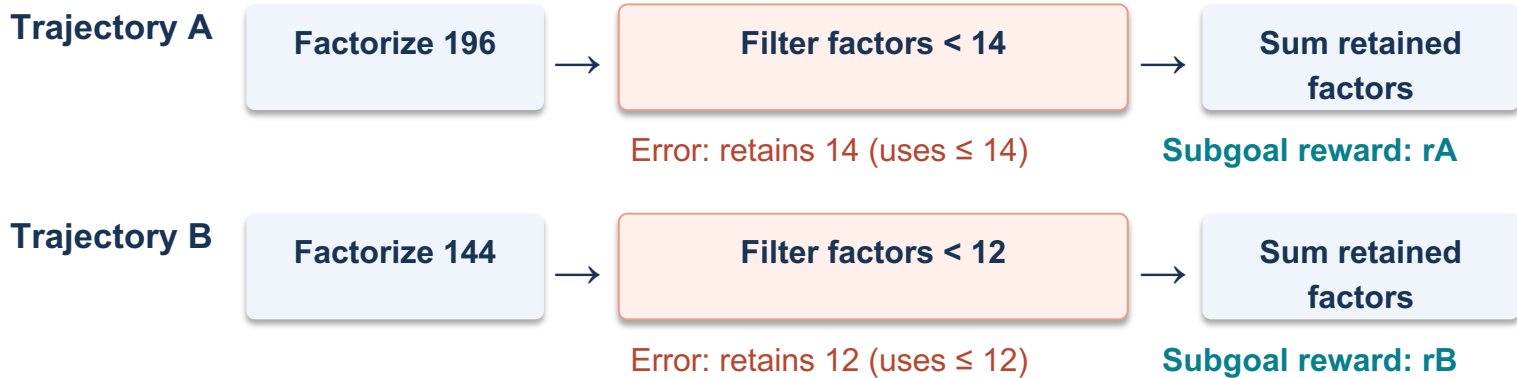
Fixed: problem, trajectory, judge, and rubric (when used). Only sampling varies.

Method	Agreement $\uparrow$ (%)	Reward Std. $\downarrow$
No-rubric PRM	71.8	0.63
RaR	82.4	0.41
SCRIBE	93.7	0.18

Protocol schematic. Results summarize repeated evaluations across the experimental inputs.

Different Trajectories, Same Error: Consistent Rewards?

Find similar subgoals with the same failure, then compare their subgoal rewards.



Same Skill + Same Failure

Strict-threshold filtering: includes the boundary.
Compare r_A and r_B for the highlighted spans.

Method	Agree ↑	Gap ↓
Direct PRM	58.6%	0.82
RaR	69.4%	0.57
SCRIBE	86.8%	0.24

Reusable Rubrics Make Reward Judgments More Consistent

Same input, repeated 10 times

Fix the problem, trajectory, rubric, and judge.

Method	Agree ↑ (%)	Std. ↓
No-rubric PRM	71.8	0.63
RaR	82.4	0.41
SCRIBE	93.7	0.18

Similar behavior, different problems

50 matched behavior groups; 2–11 subgoals each.

Method	Agree ↑ (%)	Gap ↓
Direct PRM	58.6	0.82
RaR	69.4	0.57
SCRIBE	86.8	0.24

One shared rubric gives similar reasoning behaviors a more consistent reward standard.

Improved Reasoning on Math Tasks

Model	Method	MATH500	AIME25
Qwen3-4B Instruct-2507	Base	89.1	43.3
	Process Reward Modeling	92.3	51.7
	SCRIBE (Ours)		
LLaMA 3.2-3B-Instruct	Base	40.8	1.7
	Process Reward Modeling	48.3	6.7
	SCRIBE (Ours)		

Improved Reasoning on Math Tasks

Model	Method	MATH500	AIME25
Qwen3-4B Instruct-2507	Base	89.1	43.3
	Process Reward Modeling	92.3	51.7
	SCRIBE (Ours)	95.8	63.3
LLaMA 3.2-3B-Instruct	Base	40.8	1.7
	Process Reward Modeling	48.3	6.7
	SCRIBE (Ours)	63.4	15.8

BFCL V4: Can the Model Use Tools Correctly?

Example: answering a weather question requires two dependent tool calls.

USER ASKS “What is the weather in Berkeley?”

1 Look Up the City

```
get_zipcode_by_city(  
    "Berkeley"  
)
```

Tool returns: 94710



2 Use the Returned ZIP Code

```
get_weather_by_zipcode(  
    94710  
)
```

Tool returns: weather data

What is checked? Correct tools • Correct arguments • Correct dependencies

V4 covers single-turn and multi-turn tool use, plus web search and memory.

Adapted from BFCL's official multi-step example (V3 category retained in V4).

gorilla.cs.berkeley.edu/blogs/13_bfcl_v3_multi_turn.html

Improved Reasoning on General Tool-Use Tasks

Method	MATH500	AIME25	BFCL Overall	Search	Memory	Single Live	Single Non-Live	Multi
Qwen3-4B-Instruct-2507								
Base	89.1	43.3	33.0	5.0	12.7	86.7	74.5	19.4
Process Reward Modeling	92.3	51.7	44.6	10.0	15.2	88.5	77.6	28.6
SCRIBE (Ours)	95.8	63.3						
LLaMA 3.2-3B-Instruct								
Base	40.8	1.7	21.5	0.5	5.2	82.0	58.3	3.9
Process Reward Modeling	48.3	6.7	24.8	5.0	8.3	83.8	59.6	8.2
SCRIBE (Ours)	63.4	15.8						

Improved Reasoning on General Tool-Use Tasks

Method	MATH500	AIME25	BFCL Overall	Search	Memory	Single Live	Single Non-Live	Multi
Qwen3-4B-Instruct-2507								
Base	89.1	43.3	33.0	5.0	12.7	86.7	74.5	19.4
Process Reward Modeling	92.3	51.7	44.6	10.0	15.2	88.5	77.6	28.6
SCRIBE (Ours)	95.8	63.3	51.3	12.5	15.8	90.0	80.5	33.3
LLaMA 3.2-3B-Instruct								
Base	40.8	1.7	21.5	0.5	5.2	82.0	58.3	3.9
Process Reward Modeling	48.3	6.7	24.8	5.0	8.3	83.8	59.6	8.2
SCRIBE (Ours)	63.4	15.8	30.8	8.5	11.2	85.5	62.0	13.5

Takeaway: Reuse the Standard, Not the Reference Answer

Reward what a reasoning step accomplishes with a shared, reusable skill rubric.



[Project & Slides](#)

01

A Stable Standard Across Problems

Skill prototypes make reasoning quality explicit, without a golden answer for each instance.

02

Less Reward Drift

Cross-instance agreement: 69.4% with RaR → 86.8% with SCRIBE.

03

A Better Signal for Learning

Math and tool-use gains across both evaluated model families.

Build reusable criteria for intermediate behavior—not a new grading standard for every answer.